Chapter 3 — Indefinite Integration: Chapter-wise Board Practice Paper

Maharashtra HSC · Maths Part II · Board-pattern questions with full solutions · File 3 of 3

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About This Paper

This is a chapter-wise practice paper, not an official board paper. It copies the structure of the real HSC paper (Section A MCQ + very short answers; Sections B, C, D with internal choice) but uses only Chapter 3 questions.

Source labels used on every question

Label	Meaning
PYQ (month year)	Verified: copied from the board papers in your project folder (codes J-862 Mar 2024, J-174 Jul 2024, J-312 Feb 2025, J-384 Jul 2025, J-165 Feb 2026, J-276 Jun 2026)
Textbook	A solved example or exercise question from Chapter 3 of the Std XII Part II textbook
New	Written for this paper in board style. Not from any past paper.

Marking schemes. Your project folder does not contain official board marking schemes, so the mark splits shown after each solution are *indicative*: they follow the usual HSC practice of awarding marks for the method step, the key intermediate result and the final answer. Treat them as a guide, not as official policy.

Nothing here is a "guaranteed" question. Repeats in past papers show what examiners like to ask; they do not predict the next paper.

Time and marks. Maximum marks **48** · Suggested time **1 hour 50 minutes** (the same marks-per-minute rate as the real 80-mark, 3-hour paper).

How to use. Attempt Part A (questions) fully on paper, with a timer, *before* opening Part B (solutions). In the HTML version, each solution is folded — tap to open it only after you have answered.

PART A — QUESTION PAPER

General instructions: (1) Section A: Q.1 has four MCQs of 2 marks each; Q.2 has four questions of 1 mark each. (2) Section B: attempt **any four** of Q.3–Q.8 (2 marks each). (3) Section C: attempt **any four** of Q.9–Q.14 (3 marks each). (4) Section D: attempt **any four** of Q.15–Q.21 (4 marks each). (5) For MCQs, write the option letter *and* the answer. (6) log denotes natural logarithm; c is the constant of integration.

SECTION A

Q.1. Select and write the correct answer. [8]

(i) $\int x^x(1+\log x)\,dx =$ _____

(a) $\frac{1}{2}(1+\log x)^2+c$ (b) $x^{2x}+c$ (c) $x^x\log x+c$ (d) x^x+c
[2 marks · Easy · Substitution · PYQ Feb 2025]

(ii) If $f(x)=\dfrac{\sin^{-1}x}{\sqrt{1-x^2}}$ and $g(x)=e^{\sin^{-1}x}$, then $\int f(x)\,g(x)\,dx =$ _____

(a) $e^{\sin^{-1}x}(1-\sin^{-1}x)+c$ (b) $e^{\sin^{-1}x}(\sin^{-1}x-1)+c$ (c) $e^{\sin^{-1}x}(\sin^{-1}x+1)+c$ (d) $e^{\cos^{-1}x}(\cos^{-1}x-1)+c$
[2 marks · Medium · Substitution + parts · PYQ Jun 2026]

(iii) $\displaystyle\int \frac{\sqrt{\cot x}}{\sin x\cos x}\,dx =$ _____

(a) $2\sqrt{\cot x}+c$ (b) $-2\sqrt{\cot x}+c$ (c) $\frac{1}{2}\sqrt{\cot x}+c$ (d) $\sqrt{\cot x}+c$
[2 marks · Medium · Corollary IV · Textbook Misc. Ex 3 (I)(10)]

(iv) $\displaystyle\int \frac{dx}{x^2+4x+8} =$ _____

(a) $\frac{1}{2}\tan^{-1}\left(\frac{x+2}{2}\right)+c$ (b) $\tan^{-1}\left(\frac{x+2}{2}\right)+c$ (c) $\frac{1}{4}\log\left|\frac{x+2}{x+6}\right|+c$ (d) $\frac{1}{4}\tan^{-1}\left(\frac{x+2}{4}\right)+c$
[2 marks · Easy · Completing the square · New]

Q.2. Answer the following. [4]

(i) Evaluate $\displaystyle\int \frac{1}{x^2+25}\,dx$. *[1 mark · Easy · Special integral · PYQ Mar 2024]*

(ii) Evaluate $\displaystyle\int \frac{5^x}{3^x}\,dx$. *[1 mark · Easy · a^x formula · PYQ Jul 2024]*

(iii) Evaluate $\displaystyle\int \frac{2x}{1+x^2}\,dx$. *[1 mark · Easy · f'/f · PYQ Feb 2026]*

(iv) Evaluate $\displaystyle\int \frac{1}{\sqrt{x}}\,dx$. *[1 mark · Easy · Power rule · PYQ Jul 2025]*

SECTION B — Attempt any FOUR [8]

Q.3. Evaluate $\displaystyle\int \sqrt{1+\sin 2x}\,dx$. *[2 · Easy · Trig identity · PYQ Feb 2026]*

Q.4. Evaluate $\displaystyle\int \frac{e^x(1+x)}{\sin^2(xe^x)}\,dx$. *[2 · Easy · Substitution · PYQ Jul 2024]*

Q.5. Evaluate $\displaystyle\int \frac{dx}{x+x^{-10}}$. *[2 · Medium · Substitution · PYQ Jul 2025]*

Q.6. Evaluate $\displaystyle\int x\log x\,dx$. *[2 · Easy · By parts · PYQ Jun 2026]*

Q.7. Evaluate $\displaystyle\int \frac{1}{25-9x^2}\,dx$. *[2 · Medium · Special integral · PYQ Feb 2025]*

Q.8. Evaluate $\displaystyle\int \frac{\cos x}{\sin^2 x + 4\sin x + 5}\,dx$. *[2 · Medium · Substitution + completing square · New]*

SECTION C — Attempt any FOUR [12]

Q.9. Evaluate $\displaystyle\int \frac{\sin(x+a)}{\cos(x-b)}\,dx$. *[3 · Medium · Compound angle · PYQ Jul 2024 & Jul 2025]*

Q.10. Evaluate $\displaystyle\int \frac{\sin x}{\sin 3x}\,dx$. *[3 · Medium · $\frac{1}{a+b\sin^2 x}$ type · PYQ Feb 2026]*

Q.11. Prove that $\displaystyle\int \frac{1}{a^2-x^2}\,dx = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + c$. *[3 · Easy–Medium · Proof · PYQ Mar 2024]*

Q.12. Evaluate $\displaystyle\int \frac{1}{5-4\cos x}\,dx$. *[3 · Medium · $t = \tan\frac{x}{2}$ · Textbook solved example]*

Q.13. Evaluate $\displaystyle\int \frac{x+2}{\sqrt{x^2+2x+3}}\,dx$. *[3 · Medium · $N = AD' + B$ · New]*

Q.14. Evaluate $\displaystyle\int e^{2x}\cos 3x\,dx$. *[3 · Hard · Cyclic by parts · Textbook Ex 3.3 I.10]*

SECTION D — Attempt any FOUR [16]

Q.15. Evaluate $\displaystyle\int \frac{3x^2+4x-5}{(x^2-1)(x+2)}\,dx$. *[4 · Medium · Partial fractions (i) · PYQ Feb 2026]*

Q.16. Evaluate $\displaystyle\int x^2\sin 3x\,dx$. *[4 · Medium · Repeated by parts · PYQ Jul 2024]*

Q.17. Evaluate $\displaystyle\int \frac{2}{3+2\sin^2 x+5\cos^2 x}\,dx$. *[4 · Hard · ÷cos²x, $t = \tan x$ · PYQ Jun 2026]*

Q.18. Evaluate $\displaystyle\int \frac{5e^x}{(e^x+1)(e^{2x}+9)}\,dx$. *[4 · Hard · Partial fractions (iii) · PYQ Mar 2024]*

Q.19. Evaluate $\displaystyle\int e^{\sin^{-1}x}\left(\frac{x+\sqrt{1-x^2}}{\sqrt{1-x^2}}\right)dx$. *[4 · Medium–Hard · $e^x[f+f']$ · PYQ Feb 2025]*

Q.20. Prove that $\displaystyle\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + c$. *[4 · Medium · Proof · PYQ Jul 2025]*

Q.21. Evaluate $\displaystyle\int (2x+3)\sqrt{x^2+4x+3}\,dx$. *[4 · Hard · $(px+q)\sqrt{quadratic}$ · New]*

PART B — SOLUTIONS IN BOARD ANSWER FORMAT

Attempt Part A first. Each solution is followed by the indicative mark split and examiner's notes.

Section A

Q.1 (i) — Answer: (d) $x^x + c$

Let $x^x = t$. Then $\log t = x \log x$. Differentiating w.r.t. x: $\frac{1}{t}\frac{dt}{dx} = x \cdot \frac{1}{x} + \log x = 1 + \log x$. $\therefore\ dt = x^x(1 + \log x)\,dx$.

$$\int x^x(1 + \log x)\,dx = \int dt = t + c = x^x + c. \qquad \textbf{(d)}$$

Examiner's note: recognising $\frac{d}{dx}(x^x) = x^x(1 + \log x)$ (logarithmic differentiation from Ch. 1) is the whole question. Option (a) is the trap for students who wrongly treat it as $\int f f'$.

Q.1 (ii) — Answer: (b) $e^{\sin^{-1} x}(\sin^{-1} x - 1) + c$

Put $\sin^{-1} x = t, \therefore\ \frac{dx}{\sqrt{1-x^2}} = dt$.

$$\int t\,e^t\,dt = t\,e^t - \int 1 \cdot e^t\,dt = e^t(t-1) + c = e^{\sin^{-1} x}(\sin^{-1} x - 1) + c. \qquad \textbf{(b)}$$

Examiner's note: quick check — differentiate each option; only (b) returns $\frac{\sin^{-1} x}{\sqrt{1-x^2}} e^{\sin^{-1} x}$.

Q.1 (iii) — Answer: (b) $-2\sqrt{\cot x} + c$

$\frac{\sqrt{\cot x}}{\sin x \cos x} = \frac{\sqrt{\cot x}}{\cot x \sin^2 x} = \frac{\operatorname{cosec}^2 x}{\sqrt{\cot x}}$ (multiply and divide by $\sin x$ in the denominator: $\sin x \cos x = \cot x \cdot \sin^2 x$). Put $\cot x = t$, $-\operatorname{cosec}^2 x\,dx = dt$:

$$\int \frac{-dt}{\sqrt{t}} = -2\sqrt{t} + c = -2\sqrt{\cot x} + c. \qquad \textbf{(b)}$$

Q.1 (iv) — Answer: (a) $\frac{1}{2}\tan^{-1}\left(\frac{x+2}{2}\right) + c$

$x^2 + 4x + 8 = (x+2)^2 + 2^2$. $\int \frac{dx}{(x+2)^2 + 2^2} = \frac{1}{2}\tan^{-1}\frac{x+2}{2} + c$. **(a)**

Examiner's note: (c) is what you get if you wrongly write $(x+2)^2 - 4$; (b) forgets $\frac{1}{a}$.

Q.2 — Answers

(i) $\int \frac{dx}{x^2 + 5^2} = \frac{1}{5}\tan^{-1}\frac{x}{5} + c.$

(ii) $\int \left(\frac{5}{3}\right)^x dx = \frac{(5/3)^x}{\log(5/3)} + c.$

(iii) Numerator is the derivative of the denominator: $\log(1 + x^2) + c$.

(iv) $\int x^{-1/2} dx = \frac{x^{1/2}}{1/2} + c = 2\sqrt{x} + c.$

Examiner's note: one mark each — the $+c$ and the correct constant ($\frac{1}{5}$, $\log\frac{5}{3}$) are what is checked.

Section B

Q.3 — $\int \sqrt{1+\sin 2x}\,dx$

$1+\sin 2x = \sin^2 x + \cos^2 x + 2\sin x\cos x = (\sin x+\cos x)^2$.

$$\therefore\ I = \int (\sin x+\cos x)\,dx = -\cos x + \sin x + c.$$

$$\boxed{I = \sin x - \cos x + c}$$

Indicative marks: perfect-square step 1, final 1.
Examiner's notes: the textbook takes $\sin x+\cos x \geq 0$ implicitly; you need not discuss sign. Do not write $\sqrt{1+\sin 2x} = 1+\sqrt{\sin 2x}$.

Q.4 — $\int \frac{e^x(1+x)}{\sin^2(xe^x)}\,dx$

Put $xe^x = t$. $\therefore\ (xe^x + e^x)\,dx = dt$, i.e. $e^x(1+x)\,dx = dt$.

$$I = \int \frac{dt}{\sin^2 t} = \int \operatorname{cosec}^2 t\,dt = -\cot t + c.$$

$$\boxed{I = -\cot(xe^x) + c}$$

Indicative marks: substitution with dt 1, final 1.
Examiner's notes: a frequent error is $+\cot$; $\int \operatorname{cosec}^2 = -\cot$. Jun 2026 asked the $\cos^2$ version, answer $\tan(xe^x)+c$.

Q.5 — $\int \frac{dx}{x+x^{-10}}$

$$I = \int \frac{dx}{x+\frac{1}{x^{10}}} = \int \frac{x^{10}}{x^{11}+1}dx.$$

Put $x^{11}+1 = t$. $\therefore\ 11x^{10}dx = dt$, $x^{10}dx = \frac{dt}{11}$.

$$I = \frac{1}{11}\int \frac{dt}{t} = \frac{1}{11}\log|t| + c = \boxed{\frac{1}{11}\log|x^{11}+1| + c}$$

Indicative marks: rewriting as $\frac{x^{10}}{x^{11}+1}$ 1, final 1.
Examiner's notes: forgetting the $\frac{1}{11}$ is the most common slip.

Q.6 — $\int x\log x\,dx$

By LIATE, take $\log x$ as the first function.

$$I = \log x\int x\,dx - \int\left[\frac{d}{dx}(\log x)\int x\,dx\right]dx = \frac{x^2}{2}\log x - \int \frac{1}{x}\cdot\frac{x^2}{2}dx$$

$$= \frac{x^2}{2}\log x - \frac{1}{2}\cdot\frac{x^2}{2} + c = \boxed{\frac{x^2}{2}\log x - \frac{x^2}{4} + c}$$

Indicative marks: correct by-parts formula with correct choice 1, final 1.
Examiner's notes: choosing x as first function forces you to integrate $\log x$ inside — longer and error-prone.

Q.7 — $\int \frac{dx}{25-9x^2}$

$$I = \frac{1}{9}\int \frac{dx}{\frac{25}{9}-x^2} = \frac{1}{9}\int \frac{dx}{\left(\frac{5}{3}\right)^2-x^2} = \frac{1}{9}\cdot\frac{1}{2\left(\frac{5}{3}\right)}\log\left|\frac{\frac{5}{3}+x}{\frac{5}{3}-x}\right| + c$$

$$\boxed{I = \frac{1}{30}\log\left|\frac{5+3x}{5-3x}\right| + c}$$

Indicative marks: correct standard form with $a = \frac{5}{3}$ 1, final 1.
Examiner's notes: $\frac{1}{9} \times \frac{3}{10} = \frac{1}{30}$. Writing $\frac{1}{10}$ (forgetting the 9) or $\frac{1}{2\cdot 5}$ (taking $a = 5$) are both wrong.

Q.8 — $\int \frac{\cos x}{\sin^2 x+4\sin x+5}\,dx$

Put $\sin x = t, \therefore\ \cos x\,dx = dt$.

$$I = \int \frac{dt}{t^2+4t+5} = \int \frac{dt}{(t+2)^2+1^2} = \tan^{-1}(t+2)+c = \boxed{\tan^{-1}(\sin x+2)+c}$$

Indicative marks: substitution and completing the square 1, final 1.

Section C

Q.9 — $\int \frac{\sin(x+a)}{\cos(x-b)}\,dx$

Write $x + a = (x - b) + (a + b)$.

$$\sin(x+a) = \sin[(x-b)+(a+b)] = \sin(x-b)\cos(a+b) + \cos(x-b)\sin(a+b).$$

$$\therefore\ I = \int \frac{\sin(x-b)\cos(a+b)+\cos(x-b)\sin(a+b)}{\cos(x-b)}dx = \int [\cos(a+b)\tan(x-b) + \sin(a+b)]\,dx.$$

Since $\cos(a+b)$ and $\sin(a+b)$ are constants,

$$\boxed{I = \cos(a+b)\log|\sec(x-b)| + x\sin(a+b) + c}$$

Indicative marks: angle split and expansion 1; reduction to tan + constant 1; final 1.
Examiner's notes: asked in both Jul 2024 and Jul 2025. Students lose the third mark by writing $\log|\cos(x-b)|$ with a + sign: $\int \tan u\,du = \log|\sec u| = -\log|\cos u|$.

Q.10 — $\int \frac{\sin x}{\sin 3x}\,dx$

$\sin 3x = 3\sin x - 4\sin^3 x = \sin x(3 - 4\sin^2 x)$.

$$I = \int \frac{dx}{3-4\sin^2 x}.$$

Dividing numerator and denominator by $\cos^2 x$:

$$I = \int \frac{\sec^2 x}{3\sec^2 x - 4\tan^2 x}dx = \int \frac{\sec^2 x}{3(1+\tan^2 x) - 4\tan^2 x}dx = \int \frac{\sec^2 x}{3-\tan^2 x}dx.$$

Put $\tan x = t$, $\sec^2 x\,dx = dt$:

$$I = \int \frac{dt}{(\sqrt{3})^2 - t^2} = \frac{1}{2\sqrt{3}}\log\left|\frac{\sqrt{3}+t}{\sqrt{3}-t}\right| + c = \boxed{\frac{1}{2\sqrt{3}}\log\left|\frac{\sqrt{3}+\tan x}{\sqrt{3}-\tan x}\right| + c}$$

Indicative marks: reduction to $\frac{1}{3-4\sin^2 x}$ 1; conversion to $\int \frac{dt}{3-t^2}$ 1; final 1.

Q.11 — Proof of $\int \frac{dx}{a^2-x^2}$

$$\frac{1}{a^2-x^2} = \frac{1}{(a-x)(a+x)} = \frac{A}{a-x} + \frac{B}{a+x} \;\Rightarrow\; 1 = A(a+x) + B(a-x).$$

Put $x = a$: $1 = 2aA \Rightarrow A = \frac{1}{2a}$. Put $x = -a$: $1 = 2aB \Rightarrow B = \frac{1}{2a}$.

$$\therefore \int \frac{dx}{a^2-x^2} = \frac{1}{2a}\int \frac{dx}{a-x} + \frac{1}{2a}\int \frac{dx}{a+x} = \frac{1}{2a}\left[\frac{\log|a-x|}{-1} + \log|a+x|\right] + c$$

$$= \frac{1}{2a}\left[\log|a+x| - \log|a-x|\right] + c = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + c. \qquad \textbf{Hence proved.}$$

Indicative marks: partial fractions with A, B 1; integration with the (-1) 1; combining logs 1.
Examiner's notes: the textbook also accepts writing $\frac{1}{(a-x)(a+x)} = \frac{1}{2a}\left[\frac{1}{a-x} + \frac{1}{a+x}\right]$ directly and verifying it. End with "Hence proved".

Q.12 — $\int \frac{dx}{5-4\cos x}$

Put $\tan\frac{x}{2} = t$. $\therefore\; dx = \frac{2\,dt}{1+t^2}$ and $\cos x = \frac{1-t^2}{1+t^2}$.

$$I = \int \frac{\frac{2\,dt}{1+t^2}}{5 - 4\cdot\frac{1-t^2}{1+t^2}} = \int \frac{2\,dt}{5+5t^2-4+4t^2} = \int \frac{2\,dt}{9t^2+1} = \frac{2}{9}\int \frac{dt}{t^2+\left(\frac{1}{3}\right)^2}$$

$$= \frac{2}{9}\cdot\frac{1}{\frac{1}{3}}\tan^{-1}\frac{t}{\frac{1}{3}} + c = \boxed{\frac{2}{3}\tan^{-1}\left(3\tan\frac{x}{2}\right) + c}$$

Indicative marks: substitution kit 1; reduced integral $\int \frac{2dt}{9t^2+1}$ 1; final 1.

Q.13 — $\int \frac{x+2}{\sqrt{x^2+2x+3}}\,dx$

$\frac{d}{dx}(x^2+2x+3) = 2x+2$. Write $x+2 = A(2x+2) + B$: $2A = 1$, $2A + B = 2 \Rightarrow A = \frac{1}{2}$, $B = 1$.

$$I = \frac{1}{2}\int \frac{2x+2}{\sqrt{x^2+2x+3}}dx + \int \frac{dx}{\sqrt{x^2+2x+3}} = I_1 + I_2.$$

I_1: put $x^2+2x+3 = t$: $\frac{1}{2}\int \frac{dt}{\sqrt{t}} = \sqrt{t} = \sqrt{x^2+2x+3}$.
I_2: $x^2+2x+3 = (x+1)^2 + (\sqrt{2})^2$, so $I_2 = \log\left|x+1+\sqrt{x^2+2x+3}\right|$.

$$\boxed{I = \sqrt{x^2+2x+3} + \log\left|x+1+\sqrt{x^2+2x+3}\right| + c}$$

Indicative marks: A, B 1; I_1 1; I_2 1.

Q.14 — $\int e^{2x}\cos 3x\,dx$

Take e^{2x} as the first function both times.

$$I = e^{2x}\frac{\sin 3x}{3} - \int 2e^{2x}\frac{\sin 3x}{3}dx = \frac{e^{2x}\sin 3x}{3} - \frac{2}{3}\int e^{2x}\sin 3x\,dx.$$

$$\int e^{2x}\sin 3x\,dx = e^{2x}\left(-\frac{\cos 3x}{3}\right) + \frac{2}{3}\int e^{2x}\cos 3x\,dx = -\frac{e^{2x}\cos 3x}{3} + \frac{2}{3}I.$$

$$\therefore\; I = \frac{e^{2x}\sin 3x}{3} + \frac{2e^{2x}\cos 3x}{9} - \frac{4}{9}I \;\Rightarrow\; \frac{13}{9}I = \frac{e^{2x}}{9}(3\sin 3x + 2\cos 3x).$$

$$\boxed{I = \frac{e^{2x}}{13}(2\cos 3x + 3\sin 3x) + c}$$

Indicative marks: first application 1; second application with I reappearing 1; solving for I 1.
Examiner's notes: check with the formula $\frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx)$, $a=2$, $b=3$. Note the textbook's general formula for the cosine case has sin and cos interchanged — trust the derivation.

Section D

Q.15 — $\int \frac{3x^2+4x-5}{(x^2-1)(x+2)}\,dx$

$(x^2-1)(x+2) = (x-1)(x+1)(x+2)$. Let

$$\frac{3x^2+4x-5}{(x-1)(x+1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{x+2}$$

$$\therefore\ 3x^2+4x-5 = A(x+1)(x+2) + B(x-1)(x+2) + C(x-1)(x+1).$$

- Put $x=1$: $2 = 6A \ \Rightarrow\ A = \frac{1}{3}$.
- Put $x=-1$: $-6 = -2B \ \Rightarrow\ B = 3$.
- Put $x=-2$: $-1 = 3C \ \Rightarrow\ C = -\frac{1}{3}$.

$$I = \frac{1}{3}\int \frac{dx}{x-1} + 3\int \frac{dx}{x+1} - \frac{1}{3}\int \frac{dx}{x+2}$$

$$\boxed{I = \frac{1}{3}\log|x-1| + 3\log|x+1| - \frac{1}{3}\log|x+2| + c}$$

Indicative marks: partial-fraction form and identity 1; A, B, C 2 (any two correct: 1); final integral 1.
Examiner's notes: this is solved Ex. 1 of the textbook's partial-fractions section, asked unchanged in Feb 2026. Arithmetic at $x=-2$: $3(4)+4(-2)-5 = -1$ and $(-3)(-1) = 3$.

Q.16 — $\int x^2 \sin 3x\,dx$

Take x^2 as the first function.

$$I = x^2\int \sin 3x\,dx - \int\left[\frac{d}{dx}(x^2)\int \sin 3x\,dx\right]dx = -\frac{x^2\cos 3x}{3} + \frac{2}{3}\int x\cos 3x\,dx.$$

Again by parts with x as first function:

$$\int x\cos 3x\,dx = \frac{x\sin 3x}{3} - \int \frac{\sin 3x}{3}dx = \frac{x\sin 3x}{3} + \frac{\cos 3x}{9}.$$

$$\therefore\ I = -\frac{x^2\cos 3x}{3} + \frac{2}{3}\left[\frac{x\sin 3x}{3} + \frac{\cos 3x}{9}\right] + c$$

$$\boxed{I = -\frac{x^2\cos 3x}{3} + \frac{2x\sin 3x}{9} + \frac{2\cos 3x}{27} + c}$$

Indicative marks: first application 1; second application 1; integration of $\sin 3x$, $\cos 3x$ with $\frac{1}{3}$ factors 1; final 1.
Examiner's notes: the $\frac{1}{3}$, $\frac{1}{9}$, $\frac{1}{27}$ factors are where marks are lost. Verify using the tabular rule (File 1 §9.4).

Q.17 — $\int \frac{2\,dx}{3+2\sin^2 x+5\cos^2 x}$

Dividing numerator and denominator by $\cos^2 x$:

$$I=\int \frac{2\sec^2 x}{3\sec^2 x+2\tan^2 x+5}dx=\int \frac{2\sec^2 x}{3+3\tan^2 x+2\tan^2 x+5}dx=\int \frac{2\sec^2 x}{5\tan^2 x+8}dx.$$

Put $\tan x=t$, $\sec^2 x\,dx=dt$:

$$I=2\int \frac{dt}{5t^2+8}=\frac{2}{5}\int \frac{dt}{t^2+\left(\frac{2\sqrt{2}}{\sqrt{5}}\right)^2}=\frac{2}{5}\cdot\frac{\sqrt{5}}{2\sqrt{2}}\tan^{-1}\left(\frac{\sqrt{5}\,t}{2\sqrt{2}}\right)+c.$$

$$\boxed{I=\frac{1}{\sqrt{10}}\tan^{-1}\left(\frac{\sqrt{5}\tan x}{2\sqrt{2}}\right)+c}$$

Indicative marks: division by $\cos^2 x$ and $\sec^2=1+\tan^2$ 1; substitution 1; standard form with $a=\sqrt{8/5}$ 1; final 1.
Examiner's notes: $\frac{2}{5}\cdot\frac{\sqrt{5}}{2\sqrt{2}}=\frac{\sqrt{5}}{5\sqrt{2}}=\frac{1}{\sqrt{10}}$. Equivalent forms like $\frac{1}{\sqrt{10}}\tan^{-1}\left(\frac{\sqrt{10}\tan x}{4}\right)$ are also correct.

Q.18 — $\int \frac{5e^x}{(e^x+1)(e^{2x}+9)}\,dx$

Put $e^x=t$, $\therefore\ e^x dx=dt$.

$$I=\int \frac{5\,dt}{(t+1)(t^2+9)}.\quad \text{Let } \frac{5}{(t+1)(t^2+9)}=\frac{A}{t+1}+\frac{Bt+C}{t^2+9}.$$

$$\therefore\ 5=A(t^2+9)+(Bt+C)(t+1).$$

- Put $t=-1$: $5=10A\ \Rightarrow\ A=\frac{1}{2}$.
- Coefficient of t^2: $0=A+B\ \Rightarrow\ B=-\frac{1}{2}$.
- Constant term: $5=9A+C\ \Rightarrow\ C=\frac{1}{2}$.

$$I=\frac{1}{2}\int\frac{dt}{t+1}-\frac{1}{2}\int\frac{t\,dt}{t^2+9}+\frac{1}{2}\int\frac{dt}{t^2+3^2}=\frac{1}{2}\log|t+1|-\frac{1}{4}\log(t^2+9)+\frac{1}{6}\tan^{-1}\frac{t}{3}+c$$

$$\boxed{I=\frac{1}{2}\log(e^x+1)-\frac{1}{4}\log(e^{2x}+9)+\frac{1}{6}\tan^{-1}\left(\frac{e^x}{3}\right)+c}$$

Indicative marks: substitution and correct partial-fraction form $\frac{Bt+C}{t^2+9}$ 1; constants 1; splitting the quadratic part into log and $\tan^{-1}$ 1; final 1.
Examiner's notes: writing $\frac{B}{t^2+9}$ (constant numerator) loses most marks. $\int\frac{t\,dt}{t^2+9}=\frac{1}{2}\log(t^2+9)$.

Q.19 — $\int e^{\sin^{-1}x}\left(\frac{x+\sqrt{1-x^2}}{\sqrt{1-x^2}}\right)dx$

Put $\sin^{-1}x=t$. $\therefore\ x=\sin t$, $\frac{dx}{\sqrt{1-x^2}}=dt$, and $\sqrt{1-x^2}=\sqrt{1-\sin^2 t}=\cos t$.

$$I=\int e^t(\sin t+\cos t)\,dt.$$

Let $f(t)=\sin t$; then $f'(t)=\cos t$. Using $\int e^t[f(t)+f'(t)]\,dt=e^t f(t)+c$:

$$I=e^t\sin t+c=\boxed{x\,e^{\sin^{-1}x}+c}$$

Indicative marks: substitution 1; reduced form $e^t(\sin t+\cos t)$ 1; identification of f, f' 1; final 1.
Examiner's notes: if you do not remember the $e^x[f+f']$ result, integrate $\int e^t\sin t\,dt$ by parts — the $\int e^t\cos t$ terms cancel. Write $\sin t=x$ in the final answer, not $\sin(\sin^{-1}x)$ left unsimplified.

Q.20 — Proof of $\int \sqrt{a^2-x^2}\,dx$

Let $I = \int \sqrt{a^2-x^2}\cdot 1\,dx$. Integrating by parts with $\sqrt{a^2-x^2}$ as first function:

$$I = \sqrt{a^2-x^2}\int 1\,dx - \int\left[\frac{d}{dx}\sqrt{a^2-x^2}\int 1\,dx\right]dx = x\sqrt{a^2-x^2} - \int \frac{-2x}{2\sqrt{a^2-x^2}}\cdot x\,dx$$

$$= x\sqrt{a^2-x^2} + \int \frac{x^2}{\sqrt{a^2-x^2}}dx = x\sqrt{a^2-x^2} + \int \frac{a^2-(a^2-x^2)}{\sqrt{a^2-x^2}}dx$$

$$= x\sqrt{a^2-x^2} + a^2\int \frac{dx}{\sqrt{a^2-x^2}} - \int \sqrt{a^2-x^2}\,dx = x\sqrt{a^2-x^2} + a^2\sin^{-1}\frac{x}{a} - I.$$

$$\therefore\ 2I = x\sqrt{a^2-x^2} + a^2\sin^{-1}\frac{x}{a} + c_1 \ \Rightarrow\ I = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + c. \qquad \textbf{Hence proved.}$$

Indicative marks: by-parts set-up 1; derivative of $\sqrt{a^2-x^2}$ 1; the $a^2-(a^2-x^2)$ step 1; solving for I 1. **Examiner's notes:** the step that earns the key mark is writing $x^2 = a^2-(a^2-x^2)$. The constant: $c = \frac{c_1}{2}$.

Q.21 — $\int (2x+3)\sqrt{x^2+4x+3}\,dx$

$\frac{d}{dx}(x^2+4x+3) = 2x+4$, and $2x+3 = (2x+4)-1$.

$$I = \int (2x+4)\sqrt{x^2+4x+3}\,dx - \int \sqrt{x^2+4x+3}\,dx = I_1 - I_2.$$

I_1: put $x^2+4x+3 = t$, $(2x+4)\,dx = dt$: $I_1 = \int t^{1/2}dt = \frac{2}{3}t^{3/2} = \frac{2}{3}(x^2+4x+3)^{3/2}$.
I_2: $x^2+4x+3 = (x+2)^2 - 1^2$. Using $\int \sqrt{X^2-a^2}\,dX = \frac{X}{2}\sqrt{X^2-a^2} - \frac{a^2}{2}\log\left|X+\sqrt{X^2-a^2}\right|$ with $X = x+2$, $a = 1$:

$$I_2 = \frac{x+2}{2}\sqrt{x^2+4x+3} - \frac{1}{2}\log\left|x+2+\sqrt{x^2+4x+3}\right|.$$

$$\boxed{I = \frac{2}{3}(x^2+4x+3)^{3/2} - \frac{x+2}{2}\sqrt{x^2+4x+3} + \frac{1}{2}\log\left|x+2+\sqrt{x^2+4x+3}\right| + c}$$

Indicative marks: splitting $2x+3 = D' - 1$ 1; I_1 1; completing square in I_2 1; final with correct sign of the log term 1.

Answer Key

Q	Answer
1 (i)–(iv)	(d), (b), (b), (a)
2 (i)	$\frac{1}{5}\tan^{-1}\frac{x}{5} + c$
2 (ii)	$\frac{(5/3)^x}{\log(5/3)} + c$
2 (iii)	$\log(1+x^2) + c$
2 (iv)	$2\sqrt{x} + c$
3	$\sin x - \cos x + c$
4	$-\cot(xe^x) + c$
5	$\frac{1}{11}\log\lvert x^{11}+1\rvert + c$

Q	Answer
6	$\frac{x^2}{2}\log x - \frac{x^2}{4} + c$
7	$\frac{1}{30}\log\left\lvert\frac{5+3x}{5-3x}\right\rvert + c$
8	$\tan^{-1}(\sin x + 2) + c$
9	$\cos(a+b)\log\lvert\sec(x-b)\rvert + x\sin(a+b) + c$
10	$\frac{1}{2\sqrt{3}}\log\left\lvert\frac{\sqrt{3}+\tan x}{\sqrt{3}-\tan x}\right\rvert + c$
11	Proof
12	$\frac{2}{3}\tan^{-1}\left(3\tan\frac{x}{2}\right) + c$
13	$\sqrt{x^2+2x+3} + \log\left\lvert x+1+\sqrt{x^2+2x+3}\right\rvert + c$
14	$\frac{e^{2x}}{13}(2\cos 3x + 3\sin 3x) + c$
15	$\frac{1}{3}\log\lvert x-1\rvert + 3\log\lvert x+1\rvert - \frac{1}{3}\log\lvert x+2\rvert + c$
16	$-\frac{x^2\cos 3x}{3} + \frac{2x\sin 3x}{9} + \frac{2\cos 3x}{27} + c$
17	$\frac{1}{\sqrt{10}}\tan^{-1}\left(\frac{\sqrt{5}\tan x}{2\sqrt{2}}\right) + c$
18	$\frac{1}{2}\log(e^x+1) - \frac{1}{4}\log(e^{2x}+9) + \frac{1}{6}\tan^{-1}\frac{e^x}{3} + c$
19	$x\,e^{\sin^{-1}x} + c$
20	Proof
21	$\frac{2}{3}(x^2+4x+3)^{3/2} - \frac{x+2}{2}\sqrt{x^2+4x+3} +$ $\frac{1}{2}\log\left\lvert x+2+\sqrt{x^2+4x+3}\right\rvert + c$

Formula Checklist for This Paper

Formula / method	Used in
$\int\frac{f'}{f} = \log\lvert f\rvert$, $\int\frac{f'}{\sqrt{f}} = 2\sqrt{f}$	1(iii), 2(iii), 5, 13
Substitution t = inner function	1(i), 1(ii), 4, 8, 18, 19
$\int\frac{dx}{x^2+a^2}$, $\int\frac{dx}{a^2-x^2}$, $\int\frac{dx}{\sqrt{x^2+a^2}}$	1(iv), 2(i), 7, 8, 10, 11, 13, 17, 18
Completing the square	1(iv), 8, 13, 21
$\div\cos^2 x$, $t = \tan x$	10, 17
$t = \tan\frac{x}{2}$	12
Compound angle split	9
By parts (LIATE, repeated, cyclic)	6, 14, 16, 20
$e^x[f + f']$	19
Partial fractions (types i, iii)	11, 15, 18
$\int\sqrt{x^2-a^2}$, $\int\sqrt{a^2-x^2}$	20, 21

Self-Assessment Guide

Mark your attempt using the indicative splits. Best-four rule applies in B, C and D.

Score (out of 48)	What it means	Next step
42–48	Board-ready for this chapter	Do Misc. Ex 3 (II) and (III) for speed
34–41	Solid; a few slips	List every mark lost; re-read the matching *Common Mistake* boxes in File 1
24–33	Methods known, execution weak	Redo File 2 Types 4–9 (all three levels) without notes, then retake this paper in a week
Below 24	Gaps in core methods	Work through File 1 Sections 4–10 example by example before retesting

Diagnostic by section: losing marks in Section A/B means formula recall — drill File 2 Section 1. Losing marks in Section C means set-up (choice of substitution, completing the square). Losing marks in Section D means multi-step accuracy — slow down and differentiate your answer to check.